

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
1st Semester of B. Pharm. Examination
University Theory Examination March-April 2018
BPH1002 PHARMACEUTICAL ANALYSIS I

Date: 27.03.2018, Tuesday

Time: 10.00 a.m. to 1.00 p.m.

Maximum Marks: 75

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
-

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. _____ expresses reproducibility of measurements.
[A] Accuracy
[B] Precision
[C] Sensitivity
[D] Error

2. _____ compound is used as primary standard for standardization of sodium hydroxide.
[A] Potassium hydrogen phthalate
[B] Hydrochloric acid
[C] Sulphuric acid
[D] Citric acid

3. Solution of known concentration is called _____.
[A] Primary standard
[B] Titrant
[C] Titrand
[D] Indicator

4. Titration performed without analyte is called _____.
[A] Back titration
[B] Blank titration
[C] Standardization
[D] Indirect titration

5. Which of the following is used as self-indicator _____?
 [A] KMnO_4
 [B] Starch
 [C] Iodine
 [D] AgCl
6. Which of the following salts dissolves to produce a basic aqueous solution?
 [A] NaCl
 [B] CH_3OONa
 [C] NH_4OH
 [D] $\text{CH}_3\text{COONH}_4$
7. The equilibrium expression for the ion product constant of water is _____.
 [A] $K_w = [\text{H}_3\text{O}^+][\text{OH}^-]/[\text{H}_2\text{O}]$
 [B] $K_w = [\text{H}_3\text{O}^+]^2[\text{O}_2]$
 [C] $K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$
 [D] $K_w = [\text{H}_3\text{O}^+]^2[\text{O}^{2-}]$
8. If K_{eq} is larger means _____.
 [A] Forward reaction
 [B] Reversible reaction
 [C] equilibrium condition
 [D] Incomplete reaction
9. Example of aprotic solvent used in non-aqueous titration is _____.
 [A] Acetic acid
 [B] Water
 [C] Trichlormethane
 [D] Butanamine
10. Buffer capacity of solution containing equal amount of acid and its salt is _____.
 [A] Maximum
 [B] Minimum
 [C] One half
 [D] Moderate
11. Measurement of conductance is dependent on _____.
 [A] Temperature
 [B] Solvent
 [C] Concentration of solute
 [D] All of the above

12. In potentiometric titration involve the measurement of _____
[A] Conductance
[B] Potential
[C] Concentration
[D] pH
13. In polarography _____ electrode is used.
[A] Platinum
[B] Glass
[C] Dropping mercury
[D] All of the above
14. In the gravimetric analysis crystal growth can be improved by precipitation from _____.
[A] Cold solution
[B] Dilute solution
[C] Neutral solution
[D] Basic solution
15. Which primary standard is used to standardize Iodine?
[A] Arsenic trioxide
[B] Potassium iodide
[C] Potassium permanganate
[D] Sodium thiosulphate
16. In complexometric titration, ligand and metal ion join together by _____.
[A] H-bonding
[B] Co-ordinate covalent bond
[C] Covalent bond
[D] Ionic bond
17. For the estimation of iron by redox titration, _____ is used as an indicator.
[A] Iodine
[B] Starch
[C] KMnO_4
[D] Potassium thiocyanate
18. Solution of acetic acid and sodium acetate is less acidic than solution of acetic acid due to _____.
[A] Diverse ion effect
[B] Common ion effect
[C] Equilibrium condition
[D] Uncommon ion effect

19. Following acid-base theory is applicable to non-aqueous acid-base titration_____
- [A] Lewis theory
 [B] Arrhenius theory
 [C] Ionization theory
 [D] Bronsted-Lawry theory
20. Fajan's method is based on _____.
- [A] Formation of coloured precipitation
 [B] Utilization of adsorption indicator
 [C] Formation of Complex
 [D] Formation of soluble coloured compound

SECTION – II

- Q 2** What is acid-base titration? Write a detail note on theories of acid-base indicator. Give applications of acid-base titration. **10**
- Q 3** Attempt any **TWO** of the following
- A** Describe in brief working of dropping mercury electrode. **05**
- B** What is conductometric titration? Enlist various factors affecting conductometric titration and explain any one in detail. **05**
- C** What is coprecipitation? Describe in brief different types of coprecipitation. **05**

SECTION – III

- Q 4** Attempt any **ONE** of the following
- A** Give comparative overview of Fajan's method, Volhard's method and Mohr's method. **10**
- B** What is redox titration? Enlist various types of redox titrations. Differentiate between iodimetry and iodometry titrations. Give applications of redox titration. **10**
- Q 5** Attempt any **FIVE** of the following
- A** Write a note on leveling and differentiating effect of solvents in non-aqueous titration. **05**
- B** Give application of potentiometric method of analysis. **05**
- C** What is complexometric titration? Give brief detail on masking and demasking reagents. **05**
- D** Describe different types of errors and its minimization. **05**
- E** Give principle of standardization of sodium hydroxide. **05**
- F** Discuss the principle of calcium gluconate. **05**
